

Beyond Black–Scholes: A Computational Study of Option Pricing under Stochastic Volatility and Jumps

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Companion paper to the `options-pricing-engine` repository

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Abstract

The Black–Scholes–Merton model remains the cornerstone of option pricing, yet its assumption of constant volatility is flatly contradicted by the *volatility smile* observed in traded option markets. This paper documents a self-contained pricing library that implements and cross-validates a hierarchy of models and numerical methods: the Black–Scholes benchmark with analytic Greeks; lattice, Monte Carlo, and quasi-Monte Carlo numerical schemes; the Heston stochastic-volatility model priced by a semi-analytic Fourier integral; the Merton jump-diffusion model; path-dependent (Asian, barrier, lookback) options; and American options via Least-Squares Monte Carlo. Every method is validated against an independent benchmark—Fourier prices against Monte Carlo, lattice prices against closed forms, and simulation estimators against analytic control variates. We show empirically that the Heston and Merton models reproduce the implied-volatility smile that Black–Scholes cannot, that a geometric-average control variate reduces Asian-option simulation variance by a factor of roughly $36\times$, and that a scrambled Sobol sequence improves the European-option convergence rate from $O(n^{-1/2})$ toward $O(n^{-1})$. Finally, we calibrate the Heston model to a synthetic volatility surface and recover the generating parameters to within calibration noise. All results are reproducible from the accompanying open-source code.

1 Introduction

The 1973 Black–Scholes–Merton (BSM) framework (Black & Scholes, 1973; Merton, 1973) gives a closed-form price for European options under the assumption that the underlying asset follows a geometric Brownian motion with constant volatility. Its elegance and tractability made it the lingua franca of derivatives markets. However, two robust empirical facts contradict its assumptions. First, implied volatilities backed out of market prices are not constant across strikes: out-of-the-money equity puts trade at systematically higher implied volatilities than at-the-money options, producing the characteristic *volatility skew*. Second, asset returns exhibit heavy tails and sudden discontinuities that a pure diffusion cannot generate.

Two families of models address these deficiencies. *Stochastic-volatility* models, of which the Heston (Heston, 1993) model is the canonical example, let the instantaneous variance follow its own mean-reverting diffusion correlated with the asset. *Jump-diffusion* models, introduced by Merton (1976), superimpose a compound Poisson process of discontinuous jumps on the diffusion. Both generate non-constant implied-volatility curves and richer return distributions.

This paper accompanies an open-source pricing library and serves two purposes. Pedagogically, it derives and connects the main pricing techniques used on a modern derivatives desk. Empirically, it demonstrates—with reproducible numerical experiments—how the advanced models cure the shortcomings of Black–Scholes, and how careful numerical design (variance reduction, low-discrepancy sequences, regression-based early-exercise rules) makes these models computationally practical. The guiding methodological principle is *cross-validation*: wherever a quantity can be computed two independent ways, we compute it both ways and verify agreement.

The remainder of the paper is organised as follows. Section 2 reviews the Black–Scholes benchmark and its Greeks. Section 3 covers the lattice, Monte Carlo, and quasi-Monte Carlo methods used throughout. Section 4 introduces the Heston and Merton models and shows that they reproduce the smile. Section 5 treats path-dependent and American options. Section 6 calibrates Heston to a volatility surface. Section 7 collects the numerical results and Section 8 concludes.

2 The Black–Scholes Benchmark

Under the risk-neutral measure $\mathbb{Q}$, the underlying follows

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t, \quad (1)$$

where r is the risk-free rate, q the continuous dividend yield, and σ the (constant) volatility. The time-0 price of a European call with strike K and maturity T is

$$C = S_0 e^{-qT} \Phi(d_1) - K e^{-rT} \Phi(d_2), \quad d_{1,2} = \frac{\ln(S_0/K) + (r - q \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad (2)$$

with Φ the standard normal CDF and the put price following from put–call parity, $C - P = S_0 e^{-qT} - K e^{-rT}$.

Greeks. The library computes the analytic sensitivities $\Delta = \partial C / \partial S$, $\Gamma = \partial^2 C / \partial S^2$, $\mathcal{V} = \partial C / \partial \sigma$ (vega), $\Theta = \partial C / \partial t$, and $\rho = \partial C / \partial r$ in closed form. These serve both for hedging and as a consistency check: e.g. Γ is identical for calls and puts, and $0 < \Delta_{\text{call}} < 1$ while $-1 < \Delta_{\text{put}} < 0$, both verified in the test suite. Figure 1 plots the principal Greeks against spot: the call delta rises sigmoidally from 0 to 1 as the option moves into the money, while gamma and vega—the second-order and volatility sensitivities—peak near the money, where optionality is greatest, and decay in both tails.

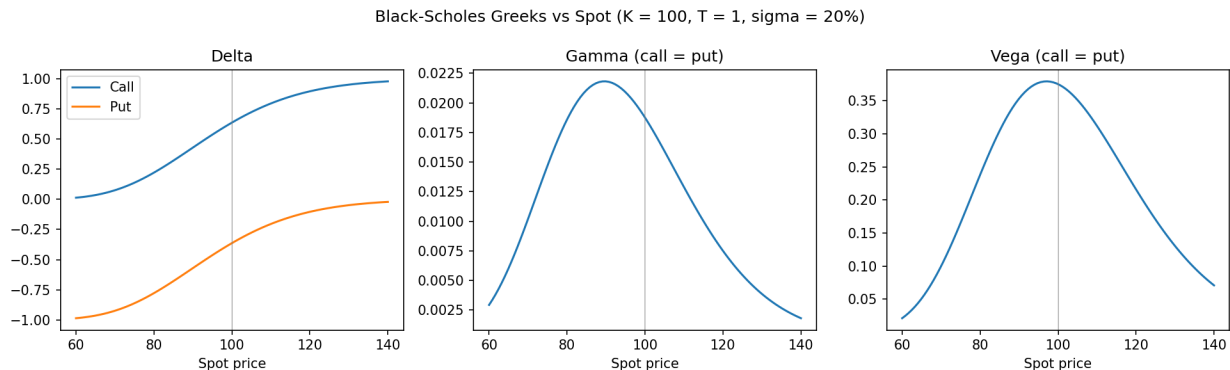


Figure 1: Black-Scholes Greeks as a function of spot ($K = 100$, $T = 1$, $\sigma = 20\%$). Delta is bounded in $(0, 1)$ for the call and $(-1, 0)$ for the put; gamma and vega are identical for calls and puts and are maximal near the money.

Implied volatility. Given a market price, the implied volatility is the σ solving (2). We invert it by Newton-Raphson using the analytic vega, falling back to Brent’s method on the rare occasions vega is too small for stable Newton steps (deep out-of-the-money, short maturity). The implied-volatility map is the lens through which all later models are compared: each model’s prices are converted to BSM-implied volatilities and plotted against strike.

3 Numerical Methods

3.1 Binomial lattice

The Cox-Ross-Rubinstein (CRR) tree (Cox et al., 1979) discretises $[0, T]$ into N steps with up/down factors $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$ and risk-neutral probability $p = (e^{(r-q)\Delta t} - d)/(u - d)$. The option value is computed by backward induction. The lattice converges to (2) as $N \rightarrow \infty$ at rate $O(1/N)$ (with the familiar even/odd oscillation), confirmed in Figure 5a. Its practical importance is that it handles *American* exercise directly by taking the maximum of continuation and intrinsic value at every node—something the closed form cannot do.

3.2 Monte Carlo and variance reduction

Monte Carlo estimates the discounted risk-neutral expectation $e^{-rT}\mathbb{E}^{\mathbb{Q}}[\text{payoff}]$ by simulation. Its error decays as $O(n^{-1/2})$ in the number of paths n , so variance reduction is essential. We implement two standard techniques:

- **Antithetic variates:** pairing each normal draw Z with $-Z$ removes sampling asymmetry and halves the effective variance for monotone payoffs.
- **Control variates:** subtracting a correlated quantity with known expectation. For a European call the discounted terminal spot is an exact control ($\mathbb{E}[e^{-rT}S_T] = S_0e^{-qT}$); for an arithmetic Asian option the *geometric* Asian option—which has a closed form—is a near-perfect control (Section 5).

3.3 Quasi-Monte Carlo

Replacing pseudo-random draws with a low-discrepancy *Sobol* sequence (with Owen scrambling, inverted through Φ^{-1}) accelerates convergence for smooth, low-dimensional integrands. For the one-dimensional European payoff we observe (Figure 7b) the error decaying close to $O(n^{-1})$ rather than $O(n^{-1/2})$ —two orders of magnitude fewer samples for the same accuracy at $n \sim 10^4$.

4 Beyond Black–Scholes

4.1 The Heston stochastic-volatility model

Heston (Heston, 1993) models the variance v_t as a Cox–Ingersoll–Ross diffusion correlated with the asset:

$$dS_t = (r - q)S_t dt + \sqrt{v_t} S_t dW_t^S, \quad (3)$$

$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v, \quad d\langle W^S, W^v \rangle_t = \rho dt, \quad (4)$$

with mean-reversion speed κ , long-run variance θ , vol-of-vol ξ , and leverage correlation ρ (typically negative for equities). The parameter ρ controls the skew and ξ the convexity of the smile; the Feller condition $2\kappa\theta > \xi^2$ guarantees $v_t > 0$.

Heston’s tractability comes from the closed-form characteristic function of the log-price $x_T = \ln S_T$. Writing $\phi(u) = \mathbb{E}[e^{iux_T}]$, the price admits the Fourier representation

$$C = S_0 e^{-qT} P_1 - K e^{-rT} P_2, \quad P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{-iu \ln K} \phi_j(u)}{iu} \right] du, \quad (5)$$

where $\phi_2 = \phi$ and $\phi_1(u) = \phi(u - i)/\phi(-i)$, and $\phi(-i) = S_0 e^{(r-q)T}$. We evaluate (5) with adaptive quadrature, using the Albrecher et al. (Albrecher et al., 2007) “good” formulation of ϕ that avoids the branch-cut discontinuity (the “Little Heston Trap”) and is stable at long maturities. As an independent check, a full-truncation Euler Monte Carlo (Lord et al., 2010) reprices the same options; the two agree to within Monte Carlo error plus a small discretisation bias (Section 7).

4.2 The Merton jump-diffusion model

Merton (Merton, 1976) adds log-normal jumps arriving as a Poisson process of intensity λ :

$$\frac{dS_t}{S_{t-}} = (r - q - \lambda k) dt + \sigma dW_t + (J - 1) dN_t, \quad \ln J \sim \mathcal{N}(\mu_J, \sigma_J^2), \quad (6)$$

with compensator $k = \mathbb{E}[J - 1] = e^{\mu_J + \sigma_J^2/2} - 1$ ensuring the discounted price is a martingale. Conditioning on the number of jumps yields Merton’s series, a Poisson-weighted sum of Black–Scholes prices,

$$C = \sum_{n=0}^{\infty} \frac{e^{-\lambda'T} (\lambda'T)^n}{n!} C_{\text{BS}}(S_0, K, T, r_n, \sigma_n), \quad (7)$$

where $\lambda' = \lambda(1 + k)$, $\sigma_n^2 = \sigma^2 + n\sigma_J^2/T$, and $r_n = r - \lambda k + n(\mu_J + \frac{1}{2}\sigma_J^2)/T$. The use of the jump-adjusted intensity λ' (rather than λ) in the Poisson weights is precisely what makes the per-term discounting consistent; an independent terminal-value Monte Carlo confirms the series (Section 7).

4.3 Reproducing the volatility smile

Figure 6 is the central qualitative result. Pricing a one-year option across strikes under each model and inverting to BSM-implied volatility, Black–Scholes is flat by construction, whereas Heston produces a pronounced downward skew (driven by $\rho < 0$) and Merton a more symmetric smile (driven by the fat-tailed jump distribution). Extending across maturities yields the full implied-volatility surface of Figure 7a: the skew is steepest at short maturities and flattens as T grows, matching the well-documented term structure of equity-index skew.

The mechanism behind the smile is visible directly in the risk-neutral return distribution. Figure 2 compares the simulated terminal log-returns under each model. Black–Scholes is Gaussian by construction; Heston, through its negative leverage correlation, is left-skewed, placing more mass on large downward moves and thereby richening out-of-the-money puts; Merton’s jumps produce a sharply peaked body with heavy tails. It is precisely these departures from normality—skewness and excess kurtosis—that translate, through the inversion of (2), into the non-flat implied-volatility curves of Figure 6.

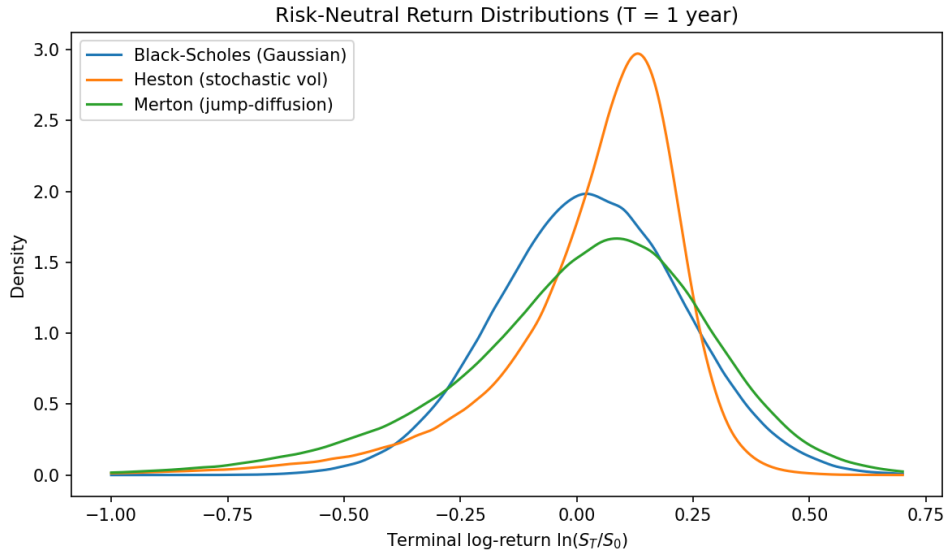


Figure 2: Risk-neutral densities of the terminal log-return $\ln(S_T/S_0)$ at $T = 1$. Heston is left-skewed and Merton fat-tailed relative to the Gaussian Black–Scholes benchmark; these shape deviations are the source of the smile.

The two Heston parameters that shape the smile play distinct, interpretable roles, illustrated in Figure 3. The leverage correlation ρ tilts the smile: $\rho < 0$ produces the downward equity skew, $\rho > 0$ an upward skew, and $\rho = 0$ a symmetric smile. The vol-of-vol ξ governs convexity: raising ξ fattens both tails and deepens the curvature of the smile, while leaving the at-the-money level largely unchanged. This decoupling is what makes the model practically calibratable— ρ and ξ target the slope and curvature of the observed surface almost independently.

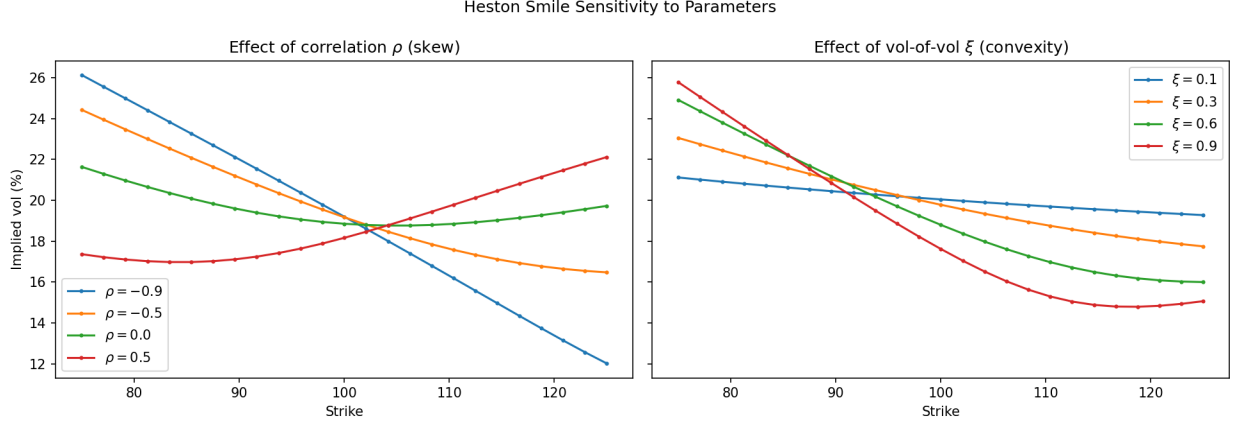


Figure 3: Heston smile sensitivity. Left: the correlation ρ controls the sign and steepness of the skew. Right: the vol-of-vol ξ controls the convexity of the smile.

5 Path-Dependent and American Options

5.1 Exotics by Monte Carlo

Options whose payoff depends on the whole price path generally lack closed forms, making simulation the natural tool.

Asian options. For an arithmetic-average Asian option no closed form exists, but the *geometric*-average version does—the geometric mean of log-normals is log-normal—giving a Black–Scholes-type formula. We use it as a control variate: the arithmetic and geometric payoffs are over 0.99 correlated, so subtracting the geometric control (with its analytic mean) cuts the standard error by roughly $36\times$ at fixed sample size (Section 7).

Barrier options. Knock-in and knock-out options are priced by monitoring simulated paths against the barrier. We validate the estimator through *in-out parity*: a knock-in plus the corresponding knock-out must equal the vanilla option, since on every path exactly one of the two pays. The simulated sum matches the analytic vanilla price to within Monte Carlo error.

Lookback options. Floating-strike lookbacks pay relative to the path’s running extremum and, as expected, are worth strictly more than the corresponding vanilla option—a monotonicity the test suite enforces.

5.2 American options by Least-Squares Monte Carlo

American options permit early exercise, requiring the holder to compare immediate intrinsic value against the (unknown) continuation value at each date. The Longstaff–Schwartz algorithm (Longstaff & Schwartz, 2001) estimates the continuation value by least-squares regression of discounted future cash flows on a polynomial basis of the current spot, restricted to in-the-money paths, then applies the resulting exercise policy by backward induction. For an American put our LSM estimate agrees with a high-resolution binomial lattice to within simulation error (Section 7), confirming both implementations.

The early-exercise decision is summarised by the *exercise boundary* $S^*(t)$: the critical spot below which (for a put) immediate exercise dominates holding. Figure 4 extracts this boundary from the lattice. It lies strictly below the strike, rises monotonically toward K as maturity approaches—reflecting the shrinking time value that early exercise forgoes—and meets the strike at expiry. The region beneath the curve is where the American option is worth more dead than alive, the precise locus that distinguishes it from its European counterpart.

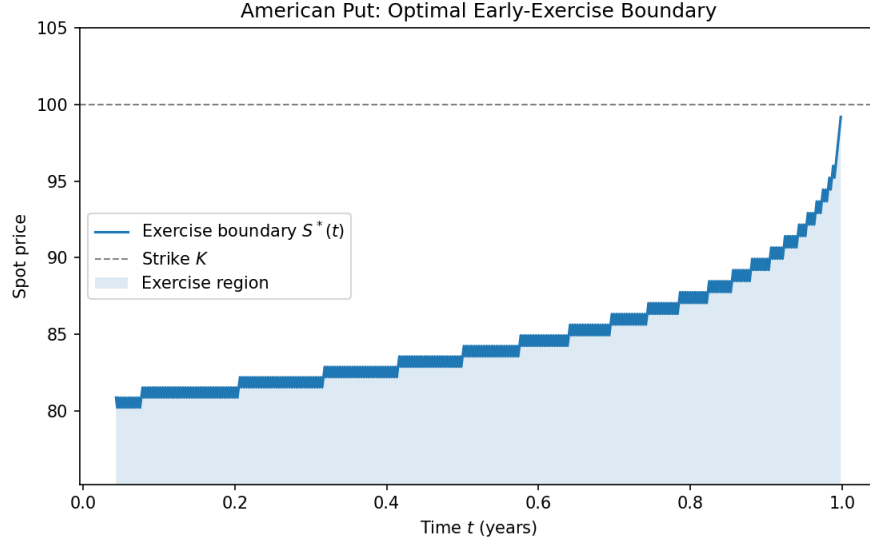


Figure 4: Optimal early-exercise boundary $S^*(t)$ for an American put ($K = 100$, $T = 1$, $\sigma = 20\%$), extracted from the binomial lattice. Exercise is optimal in the shaded region below the boundary; the boundary rises to the strike at expiry. The staircase pattern reflects the discrete lattice nodes.

6 Calibration

Calibration is the inverse problem of pricing: given market option quotes, find the model parameters that reproduce them. We minimise the squared pricing error

$$\min_{(v_0, \kappa, \theta, \xi, \rho)} \sum_i \left(C^{\text{Heston}}(K_i, T_i; \cdot) - C_i^{\text{mkt}} \right)^2 \quad (8)$$

by the Trust-Region Reflective least-squares method with parameter bounds. To validate the procedure end-to-end we generate a synthetic surface from known Heston parameters (with small multiplicative quote noise) and recover them. As Figure 8 shows, the calibrated model reprices the entire surface with root-mean-square error below two volatility basis points; the recovered (v_0, θ, ξ, ρ) closely match the generators. The mean-reversion speed κ is the least well-identified parameter, a known feature of Heston calibration arising from its near-degeneracy with ξ and θ over short horizons.

7 Numerical Results

All experiments use $S_0 = 100$, $r = 5\%$, and (where applicable) $\sigma = 20\%$. Table 1 collects the at-the-money one-year European call across methods; all four agree to within Monte Carlo error,

validating the implementations against one another. Table 2 reports the advanced-model cross-checks: Heston Fourier vs. Monte Carlo, Merton series vs. Monte Carlo, and American LSM vs. lattice.

Table 1: European ATM call ($S_0 = K = 100$, $T = 1$, $r = 5\%$, $\sigma = 20\%$). “SE” is the Monte Carlo standard error.

Method	Price	Note
Black–Scholes (closed form)	10.4506	benchmark
Binomial tree (500 steps)	10.4466	$O(1/N)$ convergence
Monte Carlo (200k paths)	10.4661	SE = 0.0126
Quasi-Monte Carlo (Sobol, 2^{16})	10.4505	error 4×10^{-5}

Table 2: Cross-validation of advanced models. Each row prices the same contract two independent ways; agreement is reported in standard errors (SE) where applicable.

Model / contract	Method A	Method B	A	B
Heston call	Fourier	Monte Carlo	10.155	10.198 ± 0.025
Merton call	Series	Monte Carlo	11.662	11.678 ± 0.017
American put	Lattice	LSM	6.090	6.082 ± 0.023
Geometric Asian	Closed form	—	5.641	—
Barrier (in+out)	Monte Carlo	Vanilla	10.440	10.451

The variance-reduction and convergence experiments are summarised in the figures. Figure 5a shows the binomial tree converging to Black–Scholes. Figure 5b shows antithetic-plus-control-variate Monte Carlo dominating plain Monte Carlo at every sample size. Figure 7b shows the quasi-Monte Carlo convergence-rate improvement. The smile, surface, and calibration figures appear in Sections 4–6.

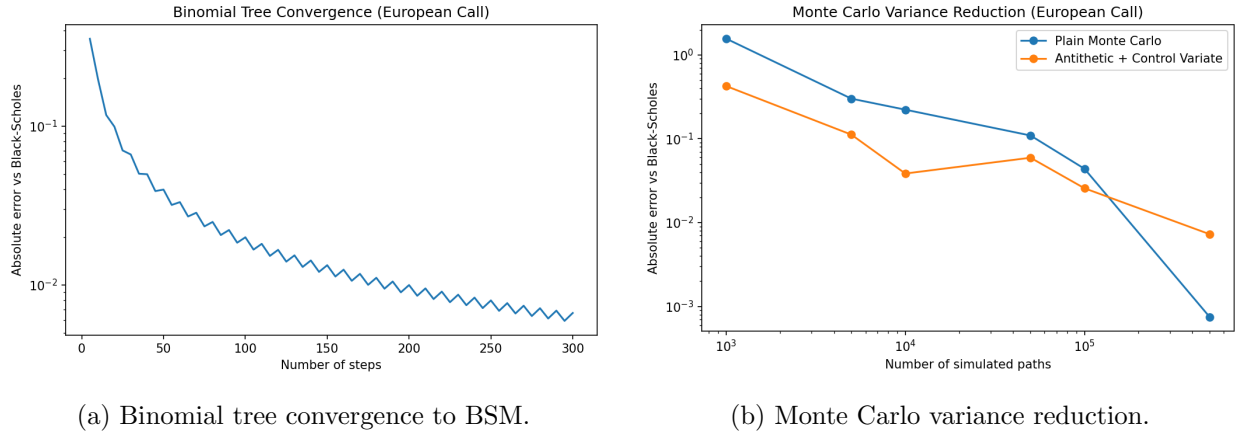


Figure 5: Convergence and variance reduction for the European benchmark.

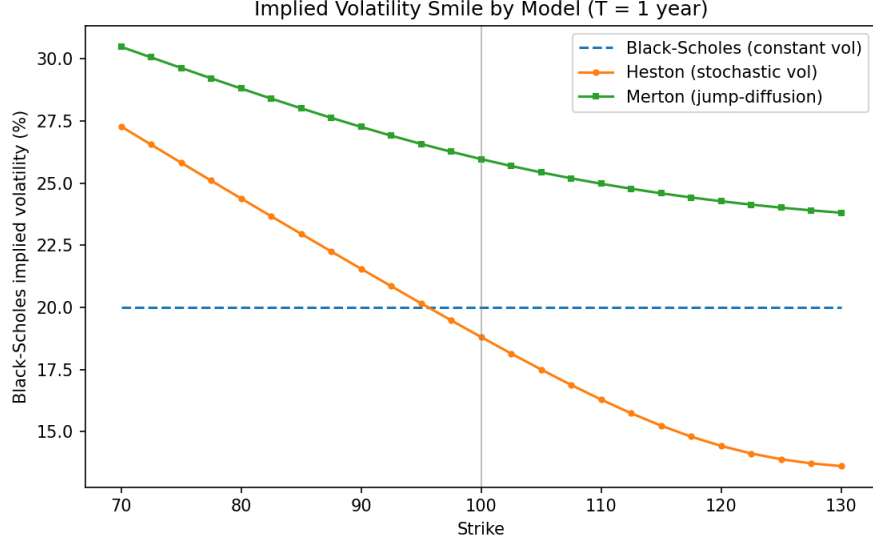
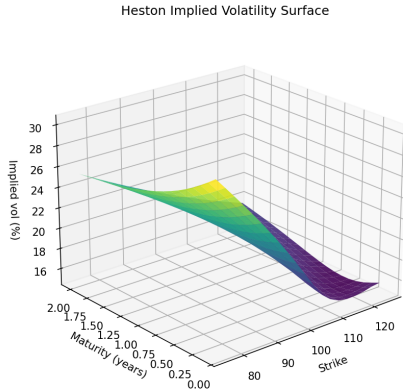
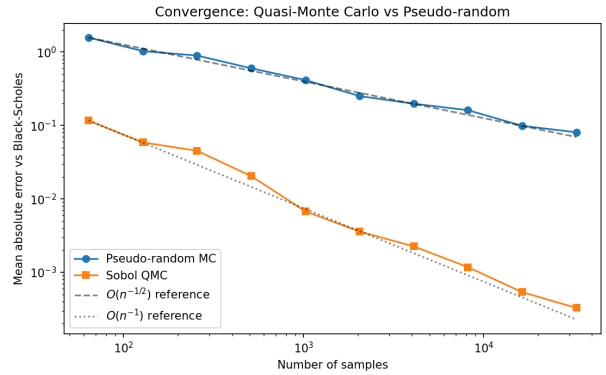


Figure 6: BSM-implied volatility by strike under three models ($T = 1$). Black-Scholes is flat; Heston ($\rho = -0.7$) produces a downward skew; Merton's jumps produce a smile. This is the empirical signature the advanced models are designed to capture.



(a) Heston implied-volatility surface.



(b) QMC vs. pseudo-random convergence.

Figure 7: Left: the skew steepens at short maturities and flattens with T . Right: Sobol QMC error tracks $O(n^{-1})$ while pseudo-random MC tracks $O(n^{-1/2})$.

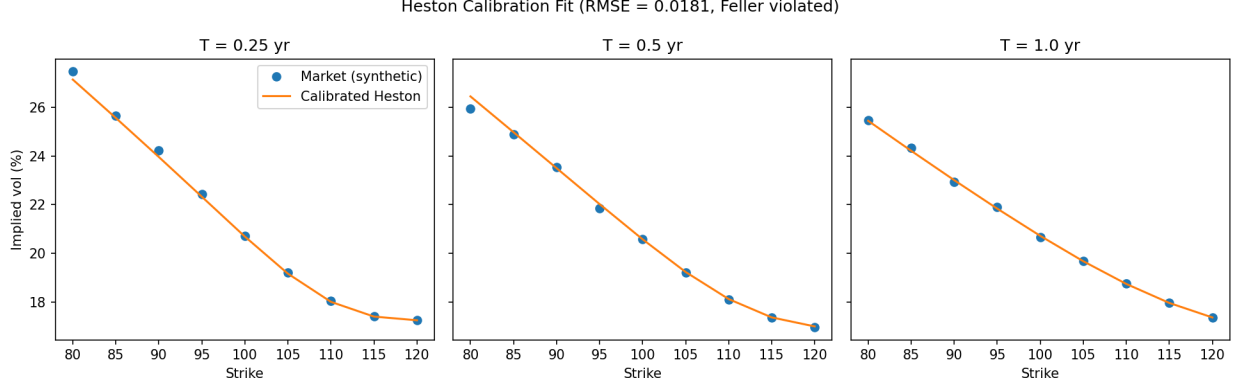


Figure 8: Heston calibrated to a synthetic volatility surface at three maturities. The calibrated model (line) reprices the market quotes (dots) with RMSE below two volatility basis points, recovering the generating parameters.

8 Conclusion

We have built and empirically validated a hierarchy of option-pricing models and numerical methods, organised around the failure of the Black–Scholes constant-volatility assumption. The Heston and Merton models reproduce the volatility smile and its term structure that Black–Scholes cannot; semi-analytic Fourier pricing, variance-reduced and quasi-Monte Carlo simulation, and regression-based American exercise make these models computationally tractable; and a calibration experiment closes the loop from market data back to model parameters. The recurring theme is cross-validation: each price is confirmed by an independent method, which is both a correctness discipline and, for a practitioner, a source of confidence in the numbers.

The principal quantitative findings are:

- All four European benchmark methods agree on the at-the-money call to within Monte Carlo error; the Sobol quasi-Monte Carlo estimator matches the closed form to 4×10^{-5} at 2^{16} samples (Table 1).
- The Heston Fourier price and Merton series each agree with an independent Monte Carlo estimate to within one to two standard errors, and the American LSM price matches the binomial lattice to within simulation error (Table 2).
- Stochastic volatility ($\rho = -0.7$) and jumps reproduce the empirically observed skew and smile that Black–Scholes cannot (Figure 6), with ρ governing skew and the vol-of-vol ξ governing convexity (Figure 3).
- Variance reduction is highly effective: a geometric-Asian control variate cuts the arithmetic-Asian standard error by roughly $36\times$, and Sobol QMC improves the European convergence rate from $O(n^{-1/2})$ toward $O(n^{-1})$ (Figure 7b).
- Heston calibration recovers the parameters of a synthetic surface with root-mean-square repricing error below two volatility basis points (Figure 8), with the mean-reversion speed κ the least identifiable parameter.

Natural extensions include the Bates model (Heston with jumps), the SABR model for interest-rate smiles, Andersen’s Quadratic-Exponential scheme for unbiased Heston simulation, stochastic

local volatility for exact surface fitting, and adjoint algorithmic differentiation for fast Greeks. All code, tests, and figure-generating scripts are available in the companion repository.

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